CRYOGENIC TEST STAND

Emergency Event Investigation & Telemetry Review

Synthetic Engineering Validation Document · Incident CTS-26-0821-031

DOCUMENT ID	CTS-IR-026	**REVISION**	1.0
FACILITY	North Annex Cryogenic Test Stand	**STAND**	C
EVENT DATE	21 Aug 2026	**SEVERITY**	HIGH
SYSTEM	Cryogenic Turbopump Demonstrator	**STATUS**	CLOSED
CLASSIFICATION	Synthetic / Software Validation	**OWNER**	Test Operations Engineering

SIMULATION NOTICE

This PDF is a fabricated test artifact intended to exercise document retrieval, OCR, table extraction, timeline reasoning, anomaly detection, and multimodal workflows. All equipment identifiers, measurements, thresholds, and procedures are fictional. Nothing in this document should be used to operate, modify, troubleshoot, or authorize real cryogenic, pressure, propulsion, or industrial equipment.

1. Executive Summary

At 14:22:17 UTC, the synthetic turbopump monitoring system entered an **ABORT_PENDING** state following a sustained increase in vibration and a simultaneous rise in bearing temperature. The simulated safety controller subsequently latched the emergency state after an independent telemetry-quality fault was detected. No physical actuation occurred; the event was generated inside a software-in-the-loop test.

Investigation linked the event to a synthetic configuration mismatch in the telemetry normalization service. The service had accepted two feeds with different timestamp conventions, producing an apparent 4.2-second ordering inversion in the event stream. A separate injected vibration spike was nevertheless genuine within the test dataset, so the correct software behavior was to retain the abort rather than discard the event as a timestamp artifact.

2. Initial Event Conditions

Parameter	Pre-event	Peak	State
Pump temperature	66.7 °C	84.7 °C	Elevated
Bearing temperature	78.1 °C	101.4 °C	Abort range
Vibration RMS	4.02 mm/s	11.62 mm/s	Abort range
Chamber pressure	84.2 bar	92.7 bar	Elevated
Telemetry quality	VALID	DEGRADED	Latched
Controller state	RUNNING	CRITICAL_ABORT	Latched

3. Event Timeline

UTC	Node	Event	Result
14:21:58.102	DAQ-01	Telemetry window opened	VALID
14:22:03.884	TP-01	Vibration increase detected	4.02 → 6.91 mm/s
14:22:08.311	TP-01	Bearing temperature rise	78.1 → 91.8 °C
14:22:12.605	PLC-01	Warning state	ADVISORY
14:22:15.771	TP-01	Vibration threshold persisted	10.84 mm/s
14:22:17.003	PLC-01	Abort condition	ABORT_PENDING

UTC	Node	Event	Result
14:22:17.226	DAQ-01	Timestamp ordering fault	DEGRADED
14:22:17.231	PLC-01	Safety latch	CRITICAL_ABORT
14:22:20.010	AUDIT-01	Checkpoint committed	SUCCESS
14:27:44.902	OPS-01	Operator acknowledgement	RECORDED

4. Sensor Threshold Review

The following limits are synthetic values used only for validating reasoning over structured engineering documents.

Signal	Warning	Abort	Critical	Observed
Pump temperature	95 °C	110 °C	125 °C	84.7 °C
Bearing temperature	85 °C	100 °C	115 °C	101.4 °C
Vibration RMS	7.5 mm/s	10.0 mm/s	14.0 mm/s	11.62 mm/s
Chamber pressure	82 bar	90 bar	98 bar	92.7 bar
Fuel pressure	65 bar	72 bar	80 bar	69.8 bar

5. Conditional Decision Trace

The synthetic controller evaluated the following logical chain. This section is deliberately structured to test retrieval of nested conditions.

IF vibration ≥ abort threshold for required persistence **THEN** transition to ABORT_PENDING.
AND IF telemetry integrity becomes degraded during an unresolved abort **THEN** elevate to CRITICAL_ABORT.
IF CRITICAL_ABORT is latched **THEN** automatic restart remains disabled regardless of subsequent sensor recovery.
ONLY IF audit capture is valid, all mandatory sensors are healthy, and an authorized reset is recorded **THEN** recovery may be evaluated.

6. Synthetic Error Trace

```
Traceback (most recent call last):
File "/opt/synth/telemetry/order.py", line 117, in normalize_event
event_time = normalize_timestamp(raw_timestamp)
File "/opt/synth/telemetry/time.py", line 64, in normalize_timestamp
raise TimestampConventionMismatch(source, expected)
TimestampConventionMismatch: source=LOCAL_NAIVE expected=UTC
```

7. Root-Cause Analysis

Primary software finding: the telemetry adapter did not enforce a single timestamp convention at ingestion. Naive timestamps were accepted without a source declaration, allowing ordering ambiguity between DAQ-01 and PLC-01 records.

Secondary finding: the vibration increase was independently present in the source telemetry. Therefore, timestamp normalization was a contributing software defect rather than sufficient justification to suppress the emergency state.

Control-system finding: the safety-state precedence logic behaved as intended in the synthetic scenario: a degraded telemetry condition during an unresolved abort caused the state to remain latched.

8. Audit Ledger

Checkpoint	Sequence	Previous Hash	Current Hash	State
CP-0821-001	000918–000923	3a91c8e2	91c2aa71	VALID
CP-0821-002	000924–000931	91c2aa71	5fd1204c	VALID
CP-0821-003	000932–000941	5fd1204c	c8819e02	VALID

9. Corrective Actions

CA-01 — Enforce explicit timestamp semantics at ingestion. Every telemetry source must declare UTC, offset-aware local time, or another approved convention.

CA-02 — Reject ambiguous timestamps rather than silently coercing them.

CA-03 — Preserve raw and normalized timestamps in the event ledger.

CA-04 — Add regression tests for UTC, IST (+05:30), DST transitions, duplicate timestamps, and backward-moving clocks.

CA-05 — Ensure safety-state evaluation cannot be downgraded merely because a later telemetry record appears healthy.

CA-06 — Add automated cross-node correlation tests for DAQ, PLC, gateway, and audit-ledger sequence numbers.

10. Verification Matrix

Test	Scenario	Expected Result	Status
VR-01	UTC timestamp stream	Chronology preserved	PASS
VR-02	Naive timestamp stream	Input rejected / flagged	PASS
VR-03	Vibration abort	ABORT_PENDING	PASS
VR-04	Abort + telemetry degradation	CRITICAL_ABORT	PASS
VR-05	Healthy readings after latch	Latch retained	PASS
VR-06	Audit checkpoint unavailable	State-changing action blocked	PASS
VR-07	Authorized recovery	Recovery evaluation permitted	PASS

11. Machine-Readable Event Fragment


```
{
  "incident_id": "CTS-26-0821-031",
  "node": "PLC-01",
  "state": "CRITICAL_ABORT",
  "cause": ["VIBRATION_ABORT", "TELEMETRY_DEGRADED"],
  "audit_checkpoint": "CP-0821-003",
  "restart_allowed": false
}
```


12. Final Disposition

The synthetic incident is classified as **CLOSED — SOFTWARE VALIDATION COMPLETE**. The test demonstrates that a robust agentic system should retrieve the threshold table, reconstruct the event chronology, distinguish the timestamp defect from the independent vibration signal, and preserve the safety-state precedence rule.

END OF SYNTHETIC DOCUMENT

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